External resource (code book) for study on SE education

Anonymized

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This document contains the thematic analysis results for a research paper on the use of generative AI (GenAI) in software engineering (SE) education. The full list of themes, subthemes, and codes can be found in the tables here. More information about this study can be found on the paper itself.

To briefly summarize this data, this analysis is based on the analysis of 25 student project reports describing how GenAI was used during the course. Using and learning to use AI to do SE was a big part of the course project. The analysis of the 25 reports resulted in 443 code occurrences across 303 quotations, where coded segments ranged from parts of sentences to full paragraphs in length.

This analysis was done partly deductively and partly inductively. Predetermined high-level categories were used to direct the coding on a general level. We analyzed the reports for data related to the SE process, and specifically for data related to the following three SE activities: (1) requirements engineering, (2) development & implementation, and (3) testing. This was done due to the course being organized around these three activities. Furthermore, within these three activities, we analyzed the data to understand the benefits and limitations of using GenAI for these activities. Thus, the data is organized into benefits and limitations for each of the three aforementioned SE activities. Past this general categorization determined in advance, we analyzed the data inductively in terms of the themes and codes that emerged within these pre-determined categories.

Table 1: Benefits of AI use in requirements engineering in the course

Subtheme / Code	n
Availability & ease of interaction	**18**
AI is good at keeping discussion going	2
AI response clarity	10
Automatic transcription	1
Speed	5
Knowledge & coverage	**12**
AI good at NFRs	1
AI has good domain knowledge	4
Good system overview	7

Table 2: Limitations of AI use in requirements engineering in the course

Subtheme / Code	n
Impacts on RE process	**4**
Need very clear, more pre-planned questions for AI	1
RE is about human communication; AI removes the whole point	2
Removes friction; removes need for critical thinking	1
No real stakeholder perspective	**24**
AI will answer equally to every question; no priorities	2
Lack of context about project	5
No understanding of real users' context-specific needs	9
Shallow / generic responses	8
Technical issues	**6**
Consistency across outputs	2
Hallucinations; false claims about project / course context	2
Response length, context window, etc. local config limitations	2
Unnatural interaction	**9**
AI has no emotions	3
Conversation with AI does not feel like natural / realistic dialogue	4
Yesman / Too optimistic	2

Table 3: Benefits of AI use in development and implementation in the course

Subtheme / Code	n
AI output quality	**35**
AI code readability	7
Architecture	2
Functionally adequate code	5
Output quality can be improved with prompting experience	10
Particularly good at smaller chunks of code	11
Learning	**25**
AI can help understand current codebase	5
AI use for information retrieval / learning	12
Raises skill floor; equalizer	2
Reviewing AI code builds own understanding	2
Weighing trade-offs builds own understanding	4
Planning advice	**10**
Planning advice	6
Second opinion	4
Productivity & speed	**50**
Can redirect own (human) energy to more demanding things when AI handles basic things	12
Faster development / productivity	11
More iterations possible after quick initial proto	5
Quickly producing initial version of system to work on	22
Psychological	**10**
Confidence	5
Feeling of responsibility over AI-produced code	5

Table 4: Limitations of AI use in development and implementation in the course

Subtheme / Code	n
Code problems	**49**
Code smells	16
Contextual fit	7
Correctness	10
Happy path bias	4
Readability	4
Security issues	8
Impact on developer / psychological	**37**
AI overreliance	6
Confidence	4
Lack of ownership of codebase	7
Lack of understanding of codebase	7
May reduce learning / own skill development	5
Skepticism about AI code	8
Need for human	**55**
AI code often needs iterative refinement immediately	10
AI code often needs refactoring (human or AI)	14
Human expertise and critical thinking required to evaluate AI suggestions and code	14
Human needs to prompt well	4
Human needs to step in when AI is unable to solve a problem at all	2
Human required in making AI code go from functional to robust/solid/good	4
Need for constant verification / validation	7
Short-term over long-term	**34**
AI does not understand full codebase	3
AI focuses on current output correctness, not overall codebase maintainability	13
Architecture problems	10
NFRs need to be prompted for	4
Settling for worse code that just works	4

Table 5: Benefits of AI use in testing in the course

Subtheme / Code	n
Diagnosis	**10**
AI code review	7
AI generates testing infra and documentation	3
Learning	**6**
Asking AI to explain its own code for testing builds own understanding	2
Learning through AI explanations of errors and error causes	4
Repair	**13**
AI fixes its own code	9
Less human effort	4

Table 6: Limitations of AI use in testing

Subtheme / Code	n
Happy path bias	**7**
AI does not account for edge cases	4
Insufficient coverage	2
Robustness has to be requested, AI doesn't automatically test code	1
Monolithic code AI produces is hard to test	**3**
Verification burden falls on human	**26**
Correct looking AI code often still has underlying problems	10
Human effort in fixing problems found in code	7
Human needs to constantly evaluate AI code and find problems	9